

2018 JC2 H2 Chemistry Prelim Practical Examination Confidential Instructions

To prepare for 7 labs of 24 students each. 3 shifts.

Apparatus list:

For each student,

Apparatus	Quantity
50 cm ³ burettes	2
Glass rod	1
Retort stand and burette clamp	2
10 cm ³ pipette	1
25 cm ³ pipette	1
Pipette filler	1
250 cm ³ graduated flask, labelled diluted FA 1	1
Dry funnel	1
Funnel	1
250 cm ³ conical flask	1
250 cm ³ DRY conical flask, labelled FA 6	1
200 cm ³ clean, dry Styrofoam cup	2
250 cm ³ beaker, labelled reaction mixture	1
50 cm ³ measuring cylinder	1
25 cm ³ measuring cylinders	1
10 cm ³ measuring cylinder	2
Stopwatch, reading to at least 0.1 s	1
Thermometer (–10 ° C to +110 ° C, at 1 °C)	1
Test-tube rack	1
Test-tubes	8 (plus extra at teacher's bench)
Test tube holder	1
Bunsen burner and lighter	1
Blue and red litmus papers	
Wooden splints	
Wash bottle containing deionised water	1
Pen for writing on glass	1
Paper towels	some
dropping pipettes	6 (plus extra at teacher's bench)
Burette clip	2
Test-tube brush	1
Large brush for cleaning conical flask	1
Filter paper	
Delivery tube (for QA)	1

Plastic gloves	1 pair
Goggles	1
Beaker for waste	1

Chemicals:

Label	Per candidate	Identity	Preparation
FA 1	100 cm ³	0.200 mol dm ⁻³ potassium iodate(V), KIO ₃	42.80 g dm ⁻³ of anhydrous KIO ₃)
FA 2	30 cm ³	1.00 mol dm ⁻³ calcium nitrate tetrahydrate, Ca(NO ₃) ₂ ·4H ₂ O	236.2 g dm ⁻³ of hydrated Ca(NO ₃) ₂ ·4H ₂ O If anhydrous calcium nitrate, Ca(NO ₃) ₂ , the solution should contain 164.1 g dm ⁻³ of the anhydrous salt
FA 3	250 cm ³	0.0400 mol dm ⁻³ sodium thiosulfate, Na ₂ S ₂ O ₃	6.324 g dm ⁻³ of anhydrous Na ₂ S ₂ O ₃ If hydrated sodium thiosulfate is used, Na ₂ S ₂ O ₃ ·5H ₂ O, the solution should contain 9.924 g dm ⁻³ of the hydrated salt
FA 4	100 cm ³	0.2 mol dm ⁻³ potassium iodide, KI	33 g dm ⁻³ of anhydrous KI
FA 5	25 cm ³	2 mol dm ⁻³ hydrochloric acid, HCl	
Starch solution	10 cm ³	2% starch solution	20 g dm ⁻³ of soluble starch
FA 7		Mg powder	0.20 ± 0.02g
FA 8	40 cm ³	1.0 mol dm ⁻³ sulfuric acid	
FA 9		Mg powder	0.20 ± 0.02g
FA 10		Mg powder	0.20 ± 0.02g
FA 11	70 cm ³	0.1 mol dm ⁻³ copper (II) sulfate	Dissolve 249.6 g of CuSO ₄ ·5H ₂ O in each dm ³ of solution.
FA12		MnSO ₄ and Zn(NO ₃) ₂	Dissolve: 35.0 g of manganese(II) sulphate, MnSO ₄ ·H ₂ O; (or 45.0 g of manganese(II) sulphate, MnSO ₄ ·4H ₂ O) 60.0 g of zinc nitrate, Zn(NO ₃) ₂ ·6H ₂ O in distilled water and make up to 1 dm ³ .

Bench reagents:

Sodium hydroxide
 Calcium hydroxide
 Aqueous ammonia
 Nitric acid
 Hydrochloric acid
 Barium nitrate
 Silver nitrate
 Aluminium foil
 Potassium permanganate
 Sulfuric acid

Bench reagents:

For each student,

Label	Identity	Preparation
Dilute sulfuric acid	1.0 mol dm ⁻³ H ₂ SO ₄	Cautiously pour 55 cm ³ of concentrated (98%) sulfuric acid into 500 cm ³ of deionised water with continuous stirring. Make the solution up to 1 dm ³ with deionised water. Care: concentrated sulfuric acid is very corrosive
Dilute nitric acid	2.0 mol dm ⁻³ HNO ₃	
Aqueous sodium hydroxide	2.0 mol dm ⁻³ NaOH	Dissolve 80.0 g of NaOH in each dm ³ of solution. Care: <i>the process of solution is exothermic and any concentrated solution is very corrosive.</i>

Aqueous calcium hydroxide	Saturated aqueous calcium hydroxide, Ca(OH)_2	Prepare fresh limewater by leaving distilled water to stand over solid calcium hydroxide for several days, shaking occasionally. Decant or filter the solution.
Aqueous barium chloride		
Aqueous silver nitrate	0.05 mol dm^{-3} silver nitrate	Dissolve 8.5 g of AgNO_3 in each dm^3 of solution.
Litmus paper (blue and red)		
KMnO_4		
Aqueous iodine		
Filter paper		